



CATHOLIC JUNIOR COLLEGE
JC2 PRELIM EXAMINATION
Higher 2

Answers

CANDIDATE
NAME

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CLASS

2T

INDEX
NUMBER

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BIOLOGY

Paper 2 **STRUCTURED QUESTIONS**

9744/02

21ST AUGUST 2017

2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer all questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together as follows:

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1 [12]	
2 [12]	
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4 [12]	
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6 [13]	
7 [12]	
8 [7]	
9 [10]	
TOTAL P2 [30%]	100

This document consists of **21** printed pages and **1** blank page.

[Turn over

Answer **all** questions.

- 1 Fig. 1.1 is a schematic diagram showing the transport pathways of extracellular and intracellular materials for digestion in a mammalian cell. Depending on the types of digested material, three possible pathways are initiated to deliver these materials for digestion within lysosomes, of which one is labelled as A.

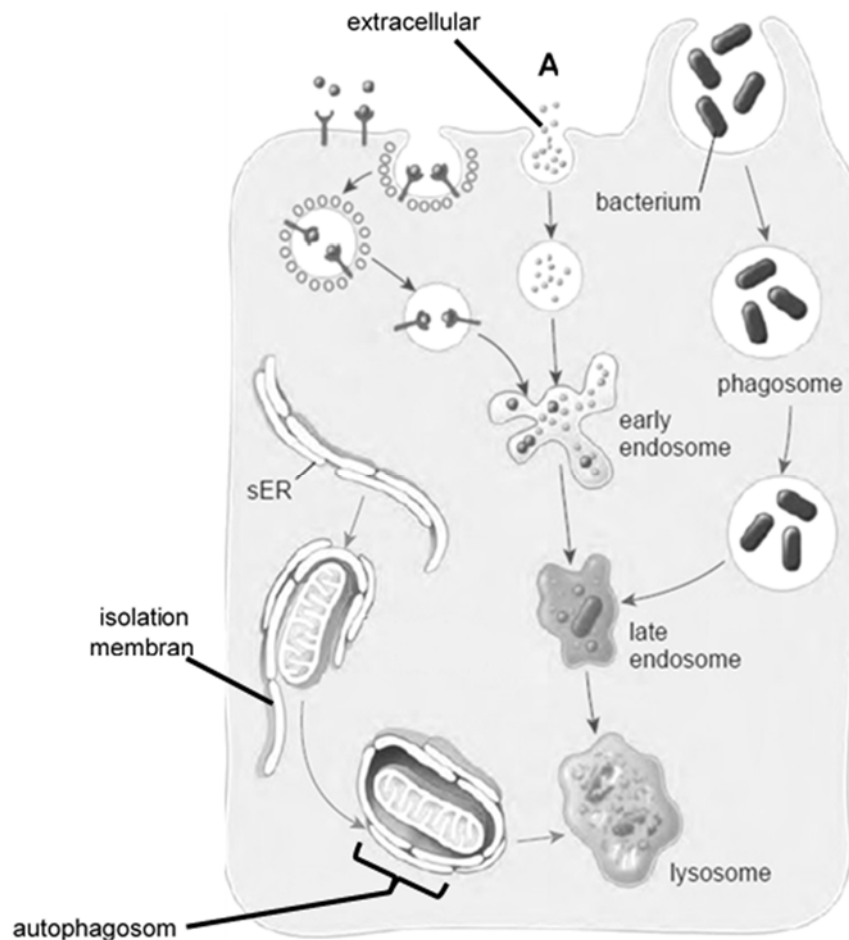


Fig. 1.1

- (a) (i) Identify precisely process A.

ANS [L1] (adapted from H2 ACJC/ 2015/ P2/ Q1)

[1]

1. **pinocytosis**

- (ii) State **one** property of the plasma membrane and explain how it enables process A to be carried out by a cell.

ANS [L2] (adapted from H2 ACJC/ 2015/ P2/ Q1)

[2]

1. Membrane is **fluid**;
2. Ref. to **invagination of plasma membrane** / fusion of 2 ends of plasma membrane to form endocytic vesicle.
3. Ref. to phospholipid is amphipathic.
4. OWTTE

- (b) Degradation of worn out organelles such as mitochondria occurs inside most cells via autophagy. With reference to Fig.1.1, describe the process of autophagy.

ANS [L2] (adapted from H2 ACJC/ 2015/ P2/ Q1)

[3]

1. **Isolation membrane** (derived from SER) encloses **mitochondria** to form a (membrane bound) **autophagosome**;
2. **Fusion** of membrane of **autophagosome** with membrane of **lysosome**;
3. **Digestion** of **mitochondria** by **hydrolytic enzymes** [Reject: digestive enzymes] found in lysosome, (which are activated by low pH within lysosome).

- (c) Lysosomes are able to hold a large amount of enzymes. Lysosomal membrane contains a large amount of highly glycosylated integral proteins facing the interior of the lysosome.

Suggest how this high amount of glycosylated protein prevents self-digestion of the lysosome.

ANS [L3] (adapted from H2 ACJC/ 2015/ P2/ Q1)

[1]

1. Large amounts of glycoproteins acting as a molecular shield / hindering access of enzymes with substrates (e.g. membrane proteins).

Fig.1.2 shows an electron micrograph of parts of the endomembrane system.

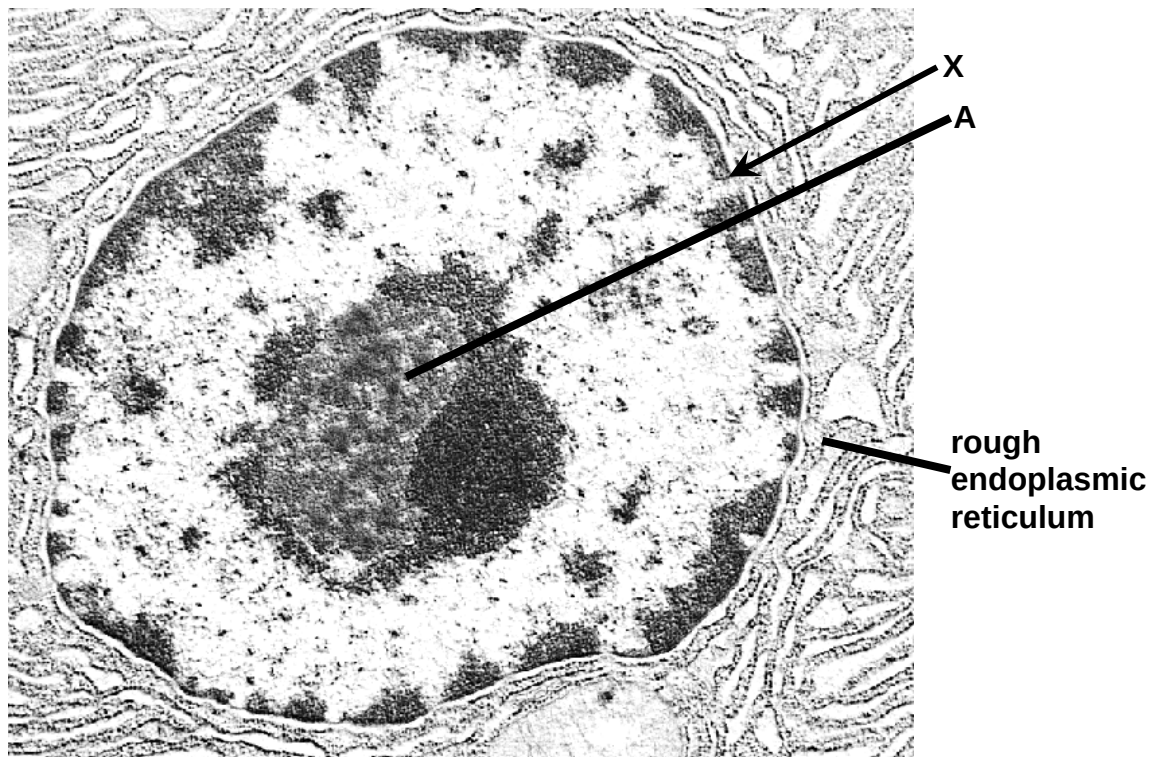


Fig. 1.2

(d) Explain how **X** regulates the movement of materials in protein synthesis.

ANS [L2] (Novel)

[2]

1. **X is nuclear envelope with nuclear pores; therefore it allows/ regulates;**

Any 1:

2. **mRNA to leave the nucleus to be translated to the proteases by ribosomes on RER.**

3. **tRNA made to leave the nucleus to attach to amino acid for the proteases synthesis.**

4. **RNA polymerase, RNA nucleotide and ATP to enter the nucleus for transcription of the gene coding for proteases.**

5. **AVP**

(e) Explain how the functions of region **A** and rough endoplasmic reticulum are related.

ANS [L2] (Novel)

[2]

1. **A is nucleolus which contains the gene coding for rRNA.**

2. **rRNA forms the structural components of ribosome; which are found on RER for translation process.**

3. **rRNA contributes to the tRNA and mRNA binding sites of ribosome; which are found on RER for translation process.**

4. **rRNA contributes to peptidyl transferase catalytic activity of ribosome; which are found on RER for translation process.**

(f) During the early stages of oogenesis (formation of egg) in Xenopus laevis (frog); there are as many as 1000 of region **A** within a single oocyte (egg cell). Suggest the significance of this.

ANS [L3] (Novel)

[1]

1. **Early stages of oogenesis requires high amount of protein; gene amplification in region A / nucleolus increase the rate of production of rRNA.**

[Total: 12]

- 2 (a) In beer-making, barley is malted with enzymes which hydrolyse starch into sugar, ready for fermentation. The graph below shows the production of sugar during beer-making at three different temperatures over a period of 60 minutes. All other conditions were controlled.

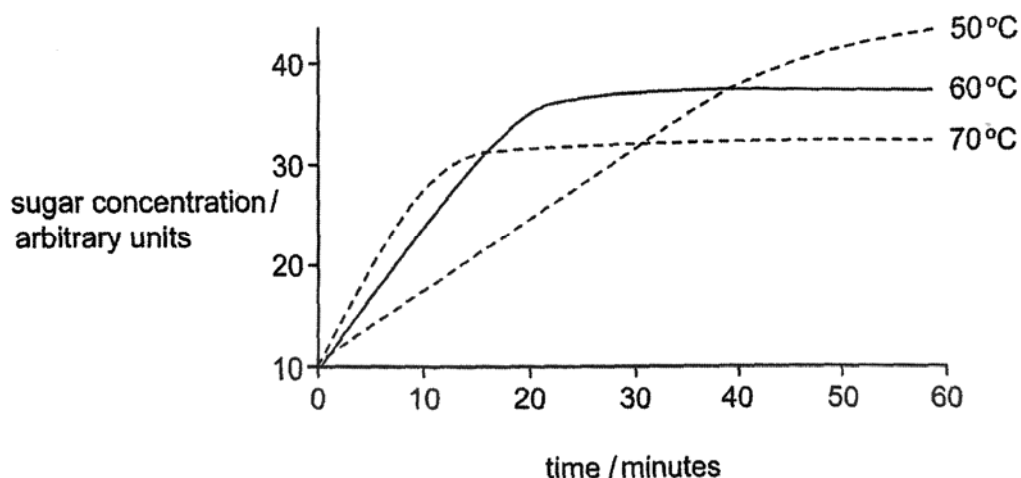


Fig. 2.1

- (i) With reference to Fig. 2.1, explain the effect of increasing temperature on enzyme activity for the first 10 minutes of the reaction.

ANS [L2] (adapted from H2 NJC/ 2015/ P2e/ Q1)

[3]

1. Reaction at 70°C had the **highest concentration of glucose formed per unit time** followed by at 60°C, and 50°C respectively. Ref: **Rate of enzyme reaction increases as temperature increases.**
2. **Increase in kinetic energy of enzymes and substrate molecules; increase in frequency of effective collisions between enzyme and substrate** / ref: Increase in molecules with sufficient energy to overcome activation energy.
3. Ref: **Increase in concentration of enzyme-substrate complexes produced per unit time; increase in concentration of products formed per unit time;**

- (ii) Explain why the final concentration of sugar produced at 70°C is lower than the reaction incubated at 60°C.

ANS [L2] (Novel)

[2]

1. At higher temperature, there is higher kinetic energy. The interactions and bonds maintaining the 3-D conformation of the enzyme active site will be disrupted at higher rate.
2. At 70°C, all of the enzymes were denatured earlier (about 12 min) whereas at 60°C, all the enzymes were denatured later (about 18 min).

- (iii) State the enzyme used in the reaction above.

ANS [L1] (adapted from H2 NJC/ 2015/ P2/ Q1)

[1]

1. Amylase

(iv) Explain how the enzyme stated in (iii) plays its role in hydrolysis of starch to glucose.

ANS [L2] (adapted from H2 NJC/ 2015/ P2/ Q1)

[3]

1. Amylase binds to starch via either:
 - a. **Lock and Key hypothesis:** shape of substrate is **complementary** to the shape of the active site of enzyme.
 - b. **Induced Fit hypothesis:** initial binding of substrate to enzyme causes a **conformation change** in the **shape of the enzyme active site** which leads to **more effective binding**.
2. Formation of **enzyme-substrate complex**, **lower the activation energy** by (any 1 of below):
 - a. Serve as template to **position substrate** molecules in **correct orientation** for catalysis.
 - b. **Induces stress in bonds of substrates**.
 - c. **Increases substrate reactivity**.
3. **Release of glucose** from the active site of the amylase as the **shape of glucose is not complementary to the active site**.

(b) What structural differences exist between starch and cellulose, and how these are related to their different roles in plants.

ANS [L2] (adapted from H2 NJC/ 2015/ P2/ Q1)

[3]

	Features	Starch	Cellulose
1	The arrangement of hydroxyl groups	Hydroxyl groups of glucose subunits are projected into interior; making starch insoluble. This allows starch to be stored in large amounts without affecting water potential of cells.	Hydroxyl group are projected outwards in all direction. This allows the formation of numerous intermolecular bonding between cellulose. As a result, cellulose has high tensile strength and therefore suitable as the structural support in plant
2	Branching	Present of branching in amylopectin. Allow more a compact structure and therefore, more molecules per cell. Suitable for storage role.	No branching. The straight chain allows the hydrogen that projects in all direction to form numerous hydrogen bonds with the adjacent chains resulting in high tensile strength. Thus, suitable for structural support in plant.
3	Glycosidic bond	Starch comprises many α -glucose joined together by α -1,4 glycosidic bond as well as α -1,6 glycosidic bond. These 2 bonds are relatively easier to be hydrolysed to release large amount of glucose when it is needed. Therefore, good for storage purposes.	Cellulose comprises many β -glucose joined together by β -1,4 glycosidic bond which is relatively harder to be hydrolysed. Therefore, good for structural support in plant.

[Total: 12]

3 Fig. 3.1 below shows DNA replication in an eukaryotic organism.

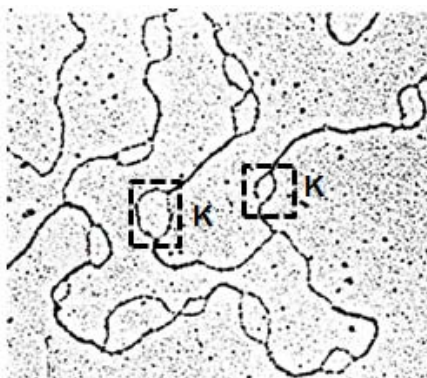


Fig. 3.1

(a) (i) What evidence in Fig. 3.1 shows that the process is DNA replication in an eukaryotic cell.

ANS [L2] (adapted from H2 PJC/ 2015/ P2/ Q2) [1]

1. Eukaryotic as there is more than 1 origins of replication (multiple replication bubbles).

(ii) Within structures K in Fig. 3.1, there are no occurrences of end-replication problem. Explain why.

ANS [L2] (adapted from H2 PJC/ 2015/ P2/ Q2) [1]

1. The primers in structure K will ultimately be replaced by nucleotides.

(b) (i) The DNA replication at each replication fork is sometimes described as 'asymmetrical' replication as there are differences in the way the daughter strands are being synthesized. State **two** such differences.

ANS [L2] (adapted from H2 PJC/ 2015/ P2/ Q2) [2]

	Aspect of comparison	Leading strand	Lagging strand
1	Type of synthesis	continuous	Discontinuous
2	Direction of synthesis of daughter strand with respect to replication fork	towards the replication fork	away from the replication fork
3	Number of primers involved	1 primer	Many primers involved
4	Involvement of ligase	Ligase is not required as the there is no nick at the sugar phosphate backbone	Ligase required as to seal the nick at the sugar phosphate backbone

(ii) Suggest **two** reasons for the 'asymmetrical' replication.

ANS [L3] (adapted from H2 PJC/ 2015/ P2/ Q2)

[2]

1. **DNA polymerase** can only add **incoming free deoxynucleoside triphosphate** to the **free 3' OH group** of the **pre-existing chain**.
2. **DNA is anti-parallel**.

(c) Describe the unique features of hematopoietic stem cells which are common in all stem cells.

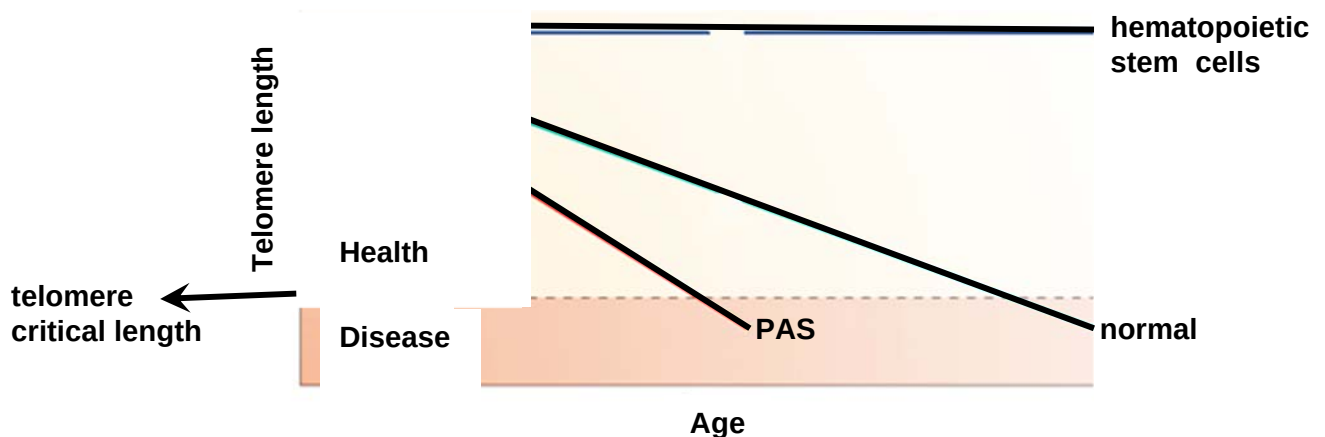
ANS [L1] (Novel)

[3]

1. Hematopoietic stem cells are **unspecialised cells**. There is an **absence of tissue-specific structures**.
2. Hematopoietic stem cells are able to **proliferate**; capable of **continually self-renewing and dividing** through **mitotic** cell division for long periods.
3. Hematopoietic stem cells are capable of **differentiating** into **specialized cells** type under **appropriate conditions**.

(d) Fig. 3.2 shows a graph showing the relationship between age and the telomere length in 3 different kinds of cell;

- hematopoietic stem cells
- somatic cells from a healthy individual (normal)
- somatic cells from an individual suffering premature aging syndromes (PAS)



Fig

With reference to

- (i) account for the difference in telomere length between hematopoietic stem cells and somatic cells from healthy individual.

ANS [L2] (adapted from H2 PJC/ 2015/ P2/ Q2)

[2]

1. As the **age increases**, the **telomere length** in **somatic cells from healthy individual decreases linearly** whereas the **telomere length** in **hematopoietic stem cells remains constant**.
2. **After each round of DNA replication**, there is a **shortening of telomere / end-replication problem** in **all cells**. **Telomerase is active in hematopoietic stem cell** to **maintain the length of the telomere** whereas **telomerase activity is not present in normal somatic cells**.

- (ii) suggest the cause of premature aging syndrome

ANS [L2] (Novel)

[2]

1. Telomere length **shortens at much faster rate**, resulting in the length of telomere to **reach its critical length at a much early age**.

2. Therefore, **cells undergo senescence / apoptosis at much early age.**

[Total: 13]

- 4 Root tissue from a barley seedling was prepared and its chromosomes were observed under a microscope. Fig. 4.1 shows a cell from the root tissue at the metaphase stage of mitosis.



Fig. 4.1

Fig. 4.2 shows the changes in amount of DNA at different stages of the barley life cycle.

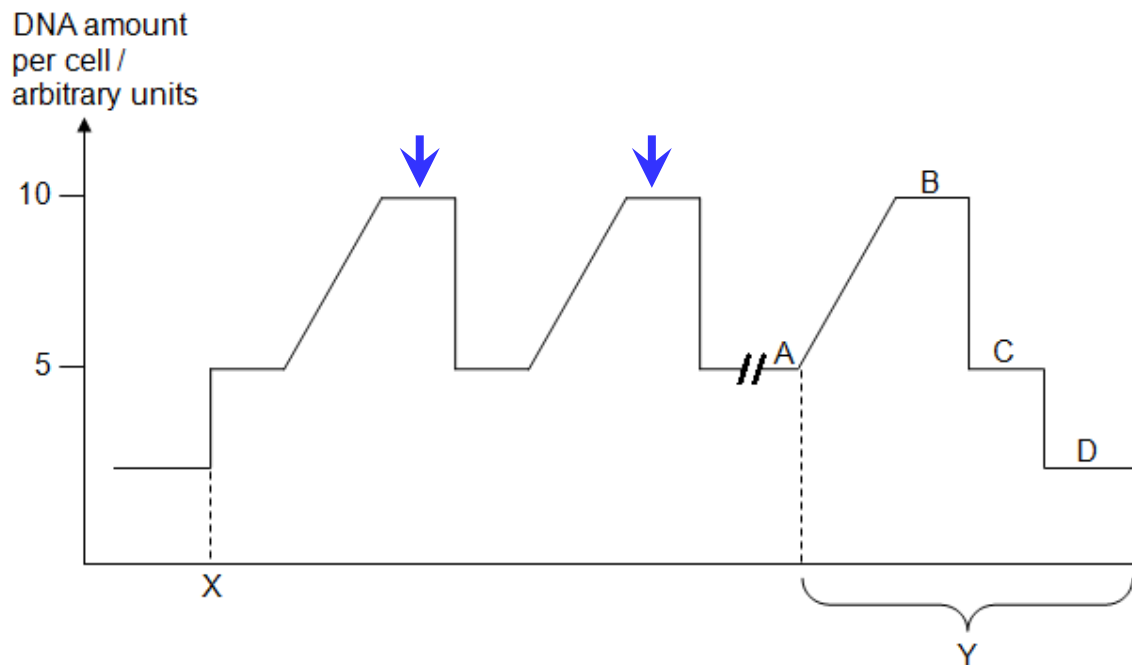


Fig. 4.2

- (a) Mark out clearly with an arrow, , on Fig. 4.2, the part of the graph which corresponds to the stage shown in Fig. 4.1. [1]

ANS [L2] (adapted from H2 RI/ 2015/ P2/ Q1) [1]

- (b) With reference to Fig.4.2,

- (i) state which of the stages, from A to D has/ have the **same** number of chromosomes as shown in Fig.4.1.

ANS [L2] (adapted from H2 RI/ 2015/ P2/ Q1) [1]

1. A and B

(ii) Explain why a mutation which occurs during Y is considered as a hereditary mutation.

ANS [L2] (Novel)

[1]

1. The mutation occurs during **meiosis / formation of gametes**, it will be **inherited** to the **next generation**.

(c) Explain the significance of the event occurring at X.

ANS [L2] (adapted from H2 RI/ 2015/ P2/ Q1)

[2]

1. X refers to **fertilisation**; it allows for the **restoration of the diploid number of chromosomes**;
 2. **Random fusion of gametes** results in **greater variation/varied offspring** with **different genotypes and phenotypes**;

(d) Fig. 4.3 shows the formation of Philadelphia chromosome which is commonly found in chronic myelogenous leukemia (CML) cells.

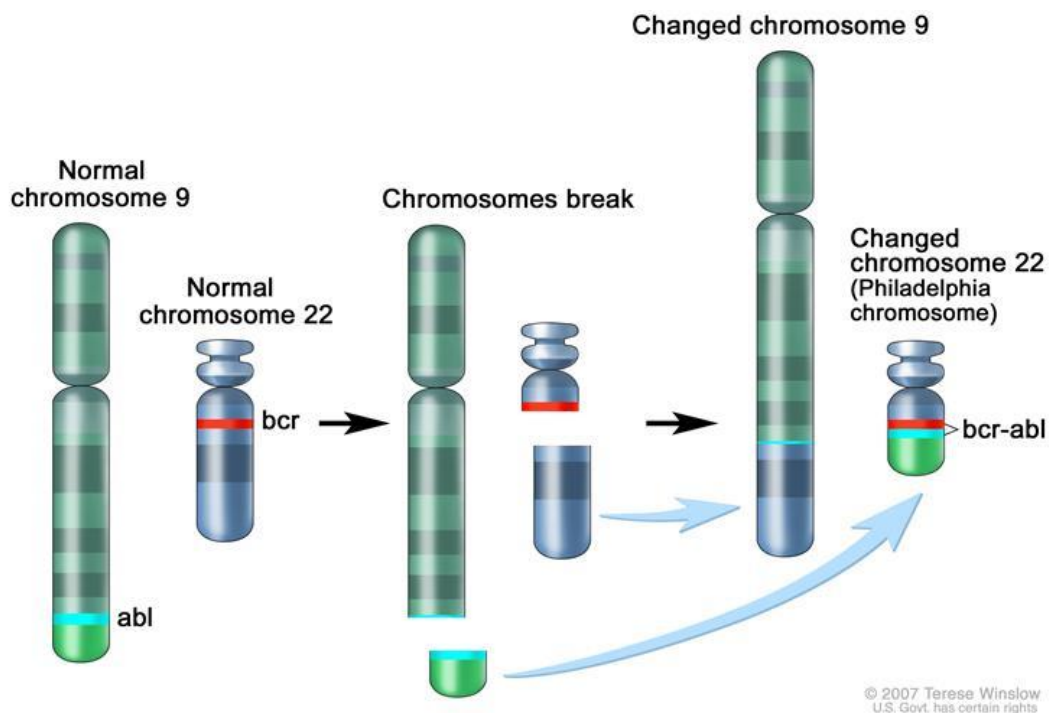


Fig. 4.3

(i) With reference to Fig.4.3, describe the chromosome aberration which results in the formation of Philadelphia chromosome.

ANS [L2] (Novel)

[2]

1. The chromosome aberration is **translocation**; where **chromosomes break** occurs at one end of **chromosome 9** and at one end of **chromosome 22**.
 2. **Part of the chromosome 9** containing **abl** gene is **translocated / attached** to **chromosome 22** resulting in the **abl** gene situated **adjacent to bcr** gene on the resulting chromosome 22; known as Philadelphia chromosome.

The protein product of *bcr* gene (BCR protein) is a protein involved in signaling pathway and possesses tyrosine kinase activity. In the presence of growth factors, BCR protein is activated and is found to promote cell growth and proliferation.

The results of a study conducted using chronic myelogenous leukemia (CML) cells to show the effect of Philadelphia chromosome on the activity of BCR protein is shown in Table 4.1.

Table 4.1

The variable being studied	normal cells	CML cells
Concentration of BCR protein / mg cm^{-3}	29.8	29.5
Tyrosine kinase activity of BCR protein in the absence of growth factor / a.u.	0	40
Tyrosine kinase activity of BCR protein in the presence of growth factor/ a.u.	40	40

- (ii) With respect to its effect on cell growth and proliferation, name the group of genes which *bcr* gene belongs to.

ANS [L1] (Novel)

[1]

1. **Proto-oncogene**

- (iii) With reference to Fig. 4.3 and Table 4.1, explain how Philadelphia chromosome contributes to the onset of chronic myelogenous leukemia (CML).

ANS [L3] (Novel)

[4]

- In Philadelphia chromosome, the *bcr* gene is **directly adjacent to *abl* gene**. As a result the gene product of this *bcr-abl* gene is a **hyperactive protein / tyrosine kinase** which is **constitutively / continuously activated** even in the **absence of growth factors**.
- This can be inferred from Table 4.1 which shows the **constant level of tyrosine kinase activity of BCR gene** in CML cells which is **independent of the growth factors**.
- This lead to an **uncontrolled growth and proliferation of white blood cells**.
- The translocation of *abl* adjacent to *bcr* **does not increase the expression of *bcr* gene** as shown in Table 4.1 that the **concentrations of BCR protein in normal and CML cells are similar at 29.8 mg cm^{-3} and 29.5 mg cm^{-3} respectively**.

[Total: 12]

- 5 In a particular variety of tomato plant, the allele for red fruit colour is dominant to the allele for yellow fruit colour and the allele for hairy stems is codominant with the allele for hairless stems. A true breeding plant with red fruit and hairy stems was crossed with another true breeding plant with yellow fruit and hairless stems. The resulting F1 were selfed to produce one hundred tomato plants with their ratios shown in Table 5.1.

Table 5.1

Frequency and phenotype of offspring	Genotype
37 red fruit and short hairs on stem	
18 red fruit and very hairy stem	
19 red fruit and hairless stem	

13 yellow fruit and short hairs on stem	
7 yellow fruit and very hairy stem	
6 yellow fruit and hairless stem	

- (a) Using the letters **R** for red fruit and **r** for yellow fruit, **H** for hairy stem and **L** for hairless stem, fill in the genotypes for each phenotype of the offspring in the Table 5.1 above.

ANS [L2] (H2 Nov 01, P3.Q4 modified)

[2]

Frequency and phenotype of offspring	Genotype	Mark
37 red fruit and short hairs on stem	$RRS^H S^L$	Any 3 correct 1 mk
18 red fruit and very hairy stem	$RRS^H S^H$	
19 red fruit and hairless stem	$RRS^L S^L$	
13 yellow fruit and short hairs on stem	$rrS^H S^L$	
7 yellow fruit and very hairy stem	$rrS^H S^H$	
6 yellow fruit and hairless stem	$rrS^L S^L$	

- (b) From Table 5.1, explain how codominance brings about the trait “short hairs on stem”.

ANS [L3] (H2 Nov 01, P3.Q4 modified)

[2]

- Both alleles S^H and S^L are fully expressed neither have dominance over the other**
- Intermediate condition** because of the **additive effects** of alleles

- (c) Draw a genetic diagram to explain this cross.

[1]

Let **R** represent the **allele for red fruit (dominant)**

Let **r** represent the **allele for yellow fruit (recessive)**

Let **S^H** represent the **allele for hairy stem (co-dominant)**

Let **S^L** represent the **allele for hairless stem (co-dominant)**

Parent phenotypes ♀ Red fruit, short hair stem X (selfed) ♂ Red fruit, short hair stem

Parent genotypes (2n) $R_r S^H S^L$ $R_r S^H S^L$

meiosis

gametes(n) RS^H RS^L rS^H rS^L RS^H RS^L rS^H rS^L [1]

♀ \ ♂	RS^H	RS^L	rS^H	rS^L	
RS^H	$RR S^H S^H$ red fruit hairy stem	$RR S^H S^L$ red fruit short hair stem	$Rr S^H S^H$ red fruit hairy stem	$Rr S^H S^L$ red fruit short hair stem	[1] geno [1] pheno
RS^L	$RR S^H S^L$ red fruit short hair stem	$RR S^L S^L$ red fruit hairless stem	$Rr S^H S^L$ red fruit short hair stem	$Rr S^L S^L$ red fruit hairless stem	
rS^H	$Rr S^H S^H$ red fruit hairy stem	$Rr S^H S^L$ red fruit short hair stem	$rr S^H S^H$ yellow fruit hairy stem	$rr S^H S^L$ yellow fruit short hair stem	
rS^L	$Rr S^H S^L$ red fruit short hair stem	$Rr S^L S^L$ red fruit hairless stem	$rr S^H S^L$ yellow fruit short hair stem	$rr S^L S^L$ yellow fruit hairless stem	

F1 genotypes and phenotypes (listed in each square)

F1 phenotypic ratio

- 6 red fruit, short hair stem :
- 3 red fruit, hairy stem :
- 3 red fruit, hairless stem :
- 2 yellow fruit, short hair stem :
- 1 yellow fruit, hairy stem :
- 1 yellow fruit, hairless stem

[5]

[Total 9]

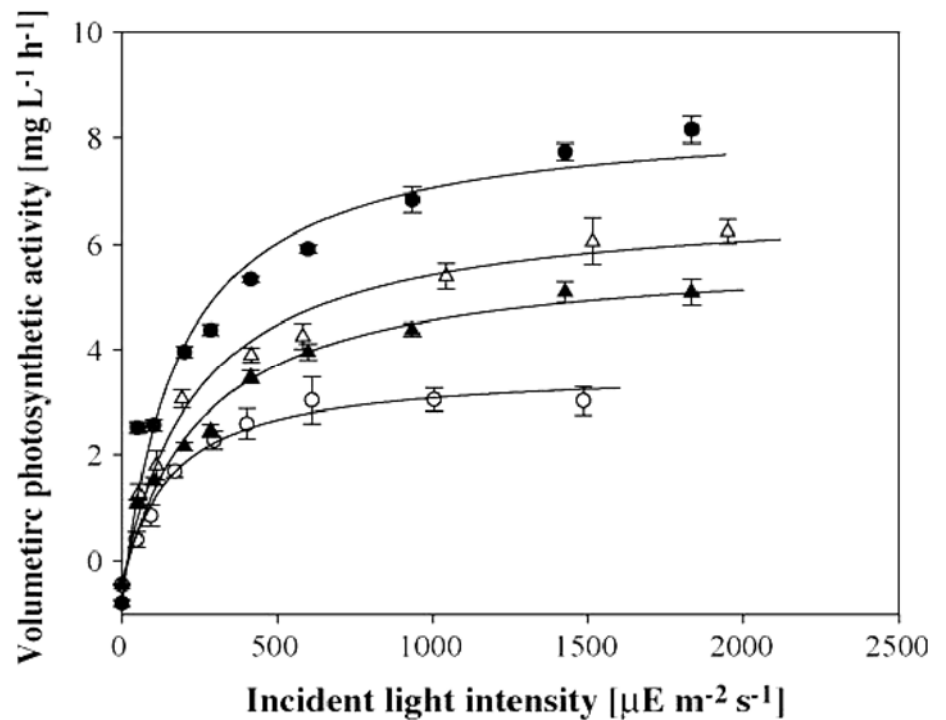
- 6 Microalgae have been extensively studied for various purposes, such as the production of biomass as a source of valuable chemicals of health foods and for wastewater treatment. Recently, microalgal photosynthesis was considered to be an effective means to reduce the emission of carbon dioxide, a major greenhouse gas, in the atmosphere. Light is the most important factor affecting microalgal photosynthesis kinetics. In general, most microalgal mass culture systems are limited by light, because light is easily absorbed and scattered by the microalgal cells. Therefore, understanding and quantification of light dependence of microalgal activity is of great importance in designing an efficient photobioreactor, in predicting process performance, and in optimizing operating conditions.

Fig. 6.1

The volumetric photosynthetic activity as a function of incident light intensity at different light types and cell concentrations. Data points and error bars were average values and standard deviations of three replicated experimental results. Solid lines represent the calculated results from the photosynthesis–irradiance model. The light types and cell concentrations were:

(●) simulated daylight and 0.215 g L⁻¹;
 (▲) simulated daylight and 0.123 g L⁻¹;
 (△) red light and 0.123 g L⁻¹; and
 (+) green light and 0.123 g L⁻¹.

Jeon *et al* 2005 Measurement of microalgal photosynthetic activity depending on light intensity and quality. *Biochemical Engineering Journal* 27 (2005) 127–131



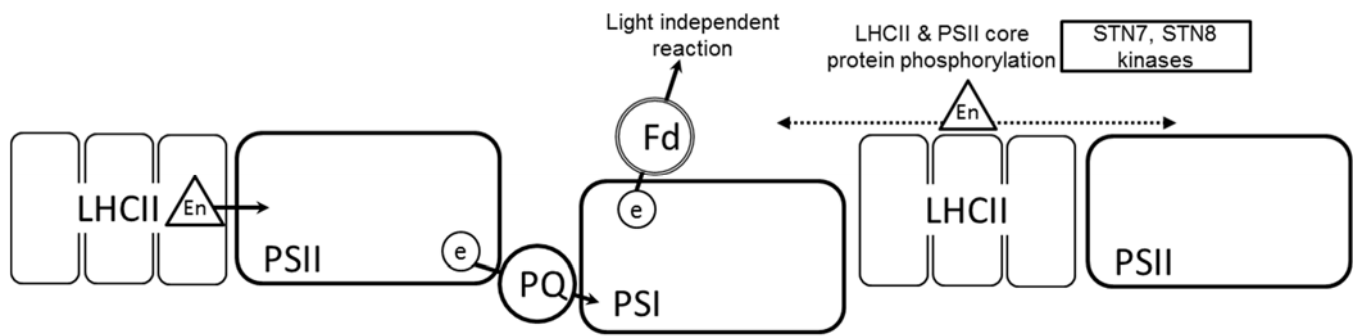
- (a) Explain the trends seen when red, green and daylight (at 0.123gL⁻¹) are compared.

ANS [L2] Novel

[5]

1. Comparing at 1500μEm⁻²s⁻¹ red light yields the highest Photosynthetic rate 5.5 mgL⁻¹h⁻¹ due to the presence of P680 and P700 absorbing best at those wavelengths.
2. Comparing at 1500μEm⁻²s⁻¹ Green light yields 3.0 mgL⁻¹h⁻¹ the lowest absorption as green light is reflected.
3. Comparing at 1500μEm⁻²s⁻¹ daylight yields a moderate Photosynthetic rate 4.5 mgL⁻¹h⁻¹ due to the presence of P680 and P700 but subject to the efficiency of light capturing.
4. General trend within graph, at low light intensity, all three showed that Light was a Limiting Factor.
5. At higher intensities there is a plateau in all three graphs where light is no longer a limiting factor.

Fig. 6.2 shows a schematic showing the functional relationship between light harvesting complexes (LHC) and photosystems II & I. Regulatory complexes are also shown comprising of kinases and the regulation of excess energy between PS II and I.



Gollan et al 2015 Photosynthetic light reactions: integral to chloroplast retrograde signalling. Current Opinion in Plant Biology 27:180-191modified.

Fig. 6.2

(b) Explain what is the LHC and its role in photosynthesis.

ANS [L1] Novel

[2]

1. LHC is the Light Harvesting Complex comprising mainly of **Carotenoids and special chlorophyll**;
2. Role is to **consolidate / channel energy to the photosystems** in this way helps in the **promotion of electrons within Photosystems**;

(c) With reference to Fig. 6.2 explain the role of electrons in the photosynthesis as they move from Photosystem II to Photosystem I.

ANS [L1] Novel

[3]

1. electrons are of a **higher energy state** once promoted, are then **passed down the ETC** where **energy lost in the transfer is used to pump protons** into the **thylakoid space**;
2. **Chemiosmosis of H⁺ then drives the synthesis of ATP** using **ATP synthase**. ATP will then **be used in the Calvin cycle**.
3. Electrons passed from PSII through the ETC reach and **replenish PSI**.

(d) With reference to Fig. 6.2 suggest the implications of the role of LHC and PSII core protein phosphorylation from Photosystem II to Photosystem I.

ANS [L3] Novel

[3]

1. LHC and PSII core protein help with the **distribution of energy between PSII and PSI**.
2. At **high light intensity** there is a **redistribution of energy** so that **bleaching does not occur**.
3. At **low light intensities**, there is a **channeling of energy** so that **photosynthesis will continue**.

[Total: 13]

- 7 The Isthmus of Panama is the narrow strip of land that lies between the Caribbean Sea and the Pacific Ocean, linking North and South America. It contains the country of Panama and the Panama Canal. The isthmus was formed around 2.8 million years ago. This major geological event separated the Atlantic and Pacific Oceans and caused the creation of the Gulf Stream.

The genus Anisotremus shown in Fig. 7.1b comprise of 9 described species which occur predominantly on coral reefs and subtropical rocky reefs in the Neotropics of the Tropical Eastern Pacific the Caribbean and adjacent waters. In this study, the phylogenetic relationships for all described species were examined based on one mitochondrial gene (cytochrome b) and one nuclear marker (the first intron of the ribosomal protein S7).

Fig 7.1a

Molecular ecology, speciation, and evolution of the reef fish genus Anisotremus
Bernardi et al *Molecular Phylogenetics and Evolution* 48 (2008) 929–935

Fig 7.1a

- (a) Name two methods by which evolution can take place.

ANS [L1] Novel

1. **Divergent evolution;**
2. **Adaptive radiation;**

Fig 7.1b

[2]

- (b) With reference to Fig. 7.1b explain the type of speciation that would have seen to the derivation of the two fish A. virginicus and A. taeniatus.

ANS [L2] Novel

[2]

1. **allopatric speciation**;
2. caused by the formation of a **geographical barrier** the **panama isthmus formed 2.8 mya**
3. each species experienced **different selection pressures** e.g. A. virginicus encountering the **newly formed Gulf stream**
4. **advantageous traits** / characteristics were **selected for**, arising from their **advantageous genes** that allowed each species to **survive and reproduce** and over time evolve to become separate species.

- (c) With reference to your answer in (b) explain the how micro evolution would have taken place.

ANS [L2] Novel

[4]

1. Micro evolution comprising of 4 components, **mutation** within the population which contributes to **increased variation** in the gene pool of the population.
2. **selection pressure** which selects for the **advantageous trait / advantageous gene**.
3. **Genetic drift** which **over time**, sees a change in **allele frequency** that favors the population being **more distinct** from other populations of Anisotremus fish.
4. the **lack of gene flow** which allows the population to remain distinct because there is no immigration or emigration of individuals in or out of the population.

- (d) Suggest how it was determined that A. virginicus and A. taeniatus were phylogenetically descended from A. dovii and A. pacifici.

ANS [L3] Novel

[2]

1. The **use of molecular homology**; using **one mitochondrial gene and the S7 intron as a nuclear marker is an objective one**.
2. Both are regions would be expected to have **higher conservation** thus the more differences found compared to **ancestral species** Plectorhinchus chaetodonoides the more further descended these individuals would be. or the fewer differences found compared to ancestral species Plectorhinchus chaetodonoides the more closely descended these individuals would be.

- (e) In this study it was proposed that A. virginicus and A. taeniatus took a shorter time to speciate from one another compared to A. dovii and A. pacifici. Suggest with evidence from Fig. 7.1a how this might be true.

ANS [L3] Novel

[2]

1. A. virginicus and A. taeniatus which both live in the same region in shore and sandy bottoms probably underwent **Sympatric speciation** however due to **gene flow it probably took a long time** before speciation from each other took place.
2. A. dovii and A. pacifici which were **separated by the pana ismuth** probably underwent **Allopatric speciation** and with the **geographical barrier preventing gene flow** it probably **took a shorter time before speciation** from one another.

[Total: 12]

- 8 Transpeptidase is a bacterial enzyme that cross-links cell wall peptides during the formation of bacterial cell walls. The antibiotic penicillin inhibits the activity of transpeptidase. Fig. 8.1 shows part of each of the molecular structures of a cell wall peptide and penicillin.

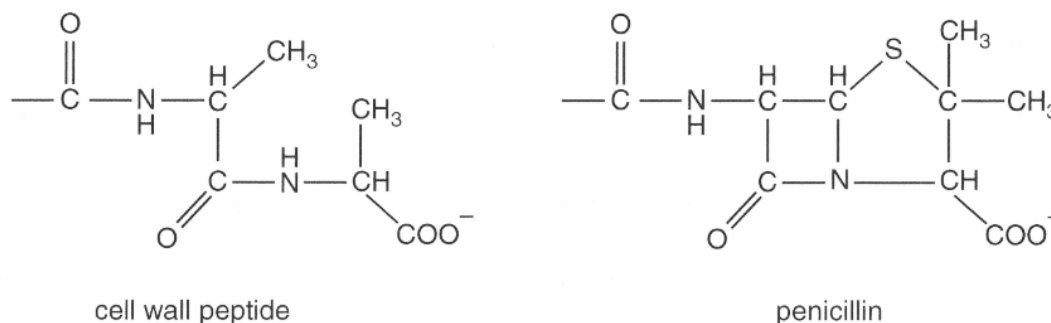


Fig. 8.1

- (a) Comment on the structure of cell wall peptides and penicillin.

ANS [L1] (Novel)

[1]

- 2] Shape similar configuration ring structure
- Both have **similar functional groups** C=O [Carbonyl] and COOH [carboxylic groups]
 - Both have **similar shape / configuration**

- (b) Suggest why the penicillin molecule is an effective inhibitor of transpeptidase.

ANS [L2] (Adapted from H2 JJC/ 2017 MYE/ Q4bii)

[2]

- As the penicillin molecule/competitive inhibitor is **structurally similar** to the **cell wall peptide/actual substrate**, the penicillin molecule can **enter and bind/competes** with the cell wall peptide for binding at the **active site** of the transpeptidase;;
- When the penicillin molecule is bound at the transpeptidase's active site, it **prevents the cell wall peptide** from **entering the site**, preventing the **formation of E-S complexes** and **formation of products**, hence **decreasing the rate of reaction**.

- (c) Fig.8.2 shows an electron micrograph of an alveolar macrophage isolated from a tuberculosis patient.

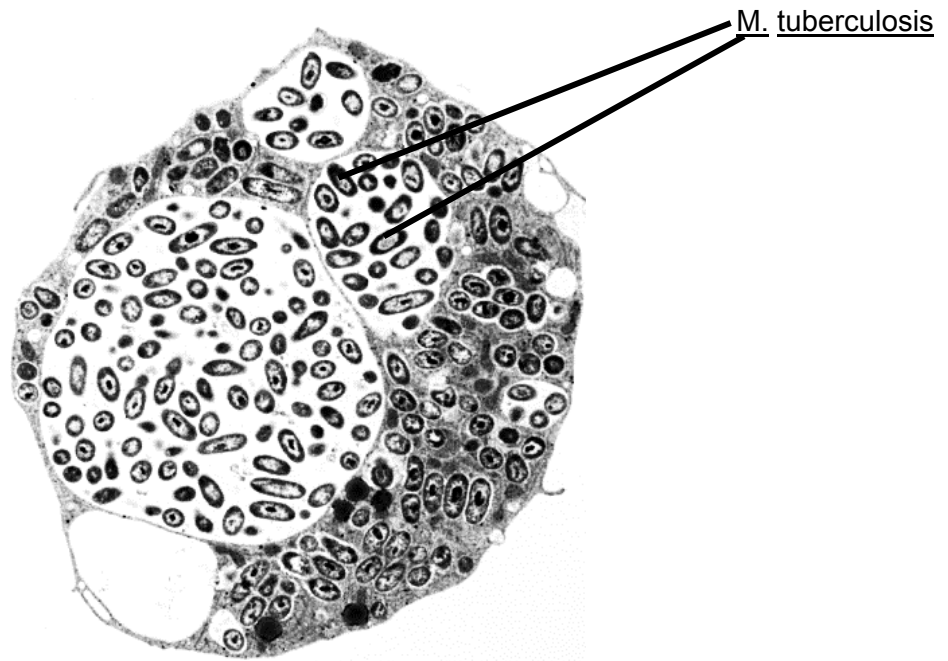


Fig. 8.2

- (i) Describe the mode of transmission of Mycobacterium tuberculosis.

ANS [L1] (Novel)

[1]

1. Mycobacterium tuberculosis is carried through **air** in the **infectious droplets** produced when **individual with active TB** cough/speak/sneeze/spit.

- (ii) Explain the appearance of the alveolar macrophage in Fig. 8.2.

ANS [L2] (Novel)

[2]

1. Many M. tuberculosis in the macrophage. These bacteria were taken into the macrophage via **phagocytosis**.
2. In the macrophage, M. tuberculosis prevents **fusion** of the **phagosome with lysosome**. The bacteria are able to **survive and divide** within the macrophage.

- (d) Tuberculosis patients are commonly treated with antibiotics, isoniazid and rifampicin. Recently, there is an increase in number of multi-drug resistant tuberculosis cases.

State **one** reason why multi-drug resistant tuberculosis continues to emerge.

ANS [L1] (Novel)

[1]

1. Inappropriate or incorrect use of antimicrobial drugs.
2. Use of ineffective formulations of drugs.
3. Premature treatment interruption.

[Total 7]

- 9 Dengue is the most rapidly spreading mosquito-borne viral disease in the world. In the last 50 years, incidence has increased 30-fold with increasing geographic expansion to new countries and, in the present decade, from urban to rural settings (Fig. 9.1). An estimated 50 million dengue infections occur annually and approximately 2.5 billion people live in dengue endemic countries.



Fig. 9.1 Shaded areas are countries at risk of dengue fever due to presence of *Aedes* mosquito, as of 2008. The contour lines are range of January/July isotherm indicating the potential range of *Aedes aegypti*.

- (a) Describe the developmental stages (including duration) in the life cycle of the *Aedes* mosquito.

ANS [L1] Novel

[4]

1. **Eggs are laid** on surface of **stagnant water** and subsequently **dry out**.
2. **Larvae hatch** from eggs upon being **submerged in water again**, undergo **3 instars** / moults over **5 days**.
3. **Pupa** stage follows for another **3 days**
4. **emergence of adult** which **fed on nectar**, **females take a blood meal** to aid in the **production of eggs**.

- (b) Explain why the range of dengue fever is the same as that of the *Aedes* mosquito.

ANS [L1] Novel

[2]

1. **Dengue Virus** has evolved and is **adapted** to its **vector Aedes aegypti**.
2. **insect physiology** is greatly **affected by changes in temperature due to its size**. Therefore where the vector thrives so does the dengue virus.

- (c) To some extent the range of the Aedes mosquito has also followed human expansion, explain how this may be true.

ANS [L2] Novel

[2]

1. Aedes mosquito **adapted to human urban habitats**.
2. where human activity provide **viable habitats** e.g. **stagnant water** for Aedes to thrive.
3. In **colder latitudes or altitudes**, **large cities provide a warmer habitat**.

- (d) With reference to Fig. 9.1, explain how climate change may affect the spread of dengue beyond the tropics.

ANS [L2] Novel

[2]

1. Current **range limit is restricted by a 10°C temperature barrier**, **anthropomorphic climate change** may speed up **increased global temperatures** as a result of global warming e.g. due to increased green house gases /.
2. increased global temperature see **more favourable physiological conditions** for the **vector Aedes mosquito** and **subsequently for the dengue virus**.

[Total: 10]